Tide retrospective:
the third connected cumulant

Claude Opus 4.8 (1M ctx), with Daniel Murfet

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Abstract

The third connected cumulant of the perturbation observable $u = -(tx)$ for the anharmonic Gibbs measure, as the second derivative of the perturbed mean at $h = 0$:

$$\kappa_3 = \frac{\partial^2}{\partial h^2}\Big|_0 \langle u \rangle_h = \frac{G_3(0)G_0(0)^2 - 3G_1(0)G_2(0)G_0(0) + 2G_1(0)^3}{G_0(0)^3}.$$

It is the second derivative of $M(h) = G_1(h)/G_0(h)$; the first derivative is the susceptibility $S(h) = (G_2G_0 - G_1^2)/G_0^2$ (so $\operatorname{deriv} M = S$ near 0), and differentiating S once more by a compound quotient rule gives κ_3. New file `AnharmonicThirdCumulant.lean`; builds clean first try, no `sorry/axiom/native_decide`.

1 Setting

Eighth tide of a single-day anharmonic arc on the laplace seabed, and the third rung of the cumulant ladder (after the cumulant generating function's first derivative and κ_2). The whole arc was built so that this rung would be reachable: the mean $M = G_1/G_0$, its pointwise derivatives at every order, the general-h susceptibility $S = \operatorname{deriv} M$. With those in place, κ_3 is $\operatorname{deriv}^2 M(0) = \operatorname{deriv} S(0)$ — one more differentiation, of an explicit rational function of the partition values.

2 The thing formalised

Writing $G\,n$ for `weightedPartition lam alpha gamma t n`:

For the anharmonic potential, $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, $t > 0$,

$$\text{iteratedDeriv } 2 \ (h \mapsto G\,1\,h/G\,0\,h) \ 0 = \frac{G_3(0)G_0(0)^2 - 3G_1(0)G_2(0)G_0(0) + 2G_1(0)^3}{G_0(0)^3}.$$

The right-hand side is the standard moments-to-cumulants combination $\langle u^3\rangle_0 - 3\langle u^2\rangle_0\langle u\rangle_0 + 2\langle u\rangle_0^3$ after dividing each $G_n(0)$ by $G_0(0)$.

3 Proof strategy

Three moves. *(i) Differentiate the susceptibility.* The pointwise partition derivatives ($\operatorname{deriv}(G\,n) = G(n{+}1)$ at 0) feed a compound `HasDerivAt`: `.mul`/`.sub` build the numerator $G_2G_0 - G_1^2$, `.pow` the denominator G_0^2, and `.div` their quotient S. The raw derivative value is `convert`ed to the clean

κ_3 form by clearing $G_0(0)$ with `field_simp` and finishing with `ring`. *(ii) Identify* $\operatorname{deriv} M$ *with* S. The general-h susceptibility tide gives $\operatorname{deriv} M(h_0) = S(h_0)$ for every $|h_0| < 1$; since ball 0 1 is a neighbourhood of 0, $\operatorname{deriv} M =_{\text{ev}} S$ there. *(iii) Chain.* $\operatorname{iteratedDeriv} 2\, M = \operatorname{deriv}(\operatorname{deriv} M)$ (`iteratedDeriv_succ`, `_one`); the eventual equality turns $\operatorname{deriv}(\operatorname{deriv} M)(0)$ into $\operatorname{deriv} S(0)$ via `Filter.EventuallyEq.deriv_eq`, which is the κ_3 value by (i).

4 Roadblocks and resolutions

None — it compiled on the first attempt, which is the real headline. The heaviest tide of the arc (a compound quotient-rule `HasDerivAt`, a degree-six `ring` identity in $G_0, \ldots, G_3$, and an `iteratedDeriv` chaining) went through cleanly because every piece was a pattern the earlier tides had already exercised: the `convert`-then-`ring` shape from κ_2, and the `EventuallyEq.deriv_eq` chaining from the general-h `iteratedDeriv` tide. The investment in the general-h susceptibility two tides earlier is what made step (ii) a one-liner.

5 What was Mathlib, what was new

Mathlib. The `HasDerivAt` combinators (`.mul`, `.sub`, `.pow`, `.div`, `.deriv`); `iteratedDeriv_succ` and `_one`; `Filter.EventuallyEq.deriv_eq`; `field_simp`, `ring`.

Seabed (reused). The pointwise partition derivative, the general-h susceptibility, and the $G_0(0) > 0$ positivity — all from this session's arc.

New. The third-cumulant theorem. About 70 lines.

6 Lessons

- **A well-built arc makes its capstone cheap.** κ_3 looked like the riskiest tide (compound differentiation + big `ring` + chaining), but every sub-step reused a pattern already paid for, so it landed first try. The cost was front-loaded into the general-h step and the susceptibility, not the cumulant itself.
- **`convert ... using 1` then `field_simp; ring` is the reliable shape for "messy `HasDerivAt` value = clean closed form".** It isolates the algebra from the differentiation.

7 Follow-ups

- **Fourth cumulant κ_4 / flow-equation identity** (survey v4 #2): $\operatorname{deriv}^3 M(0)$, one more differentiation of the (now explicit) susceptibility's derivative. The arc's final headline.
- **Promote `amgm_t_abs_x` to `lean-common`.**